

Course Title	Quantum Machine Learning (QML)		
Course Code	IT401	Credit Structure	3-0-2-4 (L-T-P-C)
Category	Elective	Semester: II/VI	Winter Semester (AY 25-26)
Program	MTech/BTech/PhD		
Prerequisites	Basics in Machine Learning (ML) and/or Deep Learning (DL), Linear Algebra		
Course Objectives/ Brief Course Description	In recent years, machine learning has expanded into virtually every domain—from military and aerospace to agriculture, manufacturing, finance, and healthcare. As ML models continue to grow in complexity, often containing billions of parameters, training them efficiently has become increasingly challenging. Quantum computing offers a promising path forward by addressing problems that are intractable for classical computing systems. Its ability to manipulate and compute over multiple Quantum bits (qubits) simultaneously enables a form of massive parallelism grounded in the principles of superposition, entanglement, and interference. Unlike classical systems that rely on sequential information processing, quantum computers leverage these quantum mechanical properties to accelerate computation in fundamentally new ways. This course introduces students to the foundational concepts of quantum computing and explores how these ideas can be applied to machine learning and deep learning. Through IBM's Qiskit (an open source SDK) tool and hands-on course projects on ML/DL, students gain experience in implementing supervised and unsupervised ML/DL algorithms using quantum and hybrid quantum–classical approaches.		
Evaluation/ Grading Policy	• Class participation - 10% • Midsem1 exam -20% • Midsem2 exam -20% • Lab and course project: - 50%.		
Course Materials/ References	1. Quantum machine learning for data scientists by Dawid Kopiczyk in arXiv, Quantum Physics, 2018 2. Quantum machine learning, What quantum computing means to data mining, book by Peter Wittek, Elsevier, 2014 3. Quantum Machine Learning with Python, book by Santanu Pattanayak, researcher Qualcomm corp, Apress, 2021 4. Quantum machine learning: A classical perspective, Proceedings Royal Society, A474, 2017 5. Quantum machine learning MOOC, created by Peter Wittek from University of Toronto, 2019 (video lectures) 6. Introduction to Quantum computing, NPTEL (video lectures)		

	7. Quantum machine learning, Qiskit Global Summer school, 2021 (video lectures)
Detailed Course Content	Difference between classical and quantum machine learning, Brief introduction to quantum applications, vector spaces, tensor products and qubits, Superposition, entanglement, interference, quantum gates, Quantum circuits, density matrix, Variational quantum circuit Grover search algorithm, Quantum Fourier transform Quantum encoding (basis, amplitude, angle, higher order encoding), quantum measurement, Quantum Eigen solver, quantum approximation optimization algorithm (QAOA) Variational quantum algorithms, noisy intermediate scale quantum (NISQ) computing, Quantum support vector machine Quantum k-means clustering Quantum neural networks (QNN). Quantum classical hybrid classification quantum classical hybrid time series analysis Quantum classical hybrid super-resolution.

Course Outcome:

CO1: Understand the basics in quantum computing and machine learning

CO2: Analyse the quantum circuits operations

CO3: Identify problems in machine learning to solve by using quantum approach

CO4: Test the algorithms in labs and carry out projects using QISKIT.

POs-COs Matrix:

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X					X	X		X